WORKED SOLUTION

STEP III 1988, Question 10

Pure • Pursuit Curves

STEP III 1988, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q10

This question has a full guided treatment in *Beyond STEP*, Section 8.5, *Greyhounds Chasing Greyhounds*: https://beyondstep.keithmcnulty.org/ch08/pursuit.html

The key idea

Don't try to track four moving dogs separately. The set-up is unchanged by a quarter-turn about the centre, so the dogs always stay at the corners of a square centred at O: it shrinks and rotates, but it stays a square. That fixes the angle between A's velocity and the line AO, and a fixed angle between velocity and radius is exactly what produces an equiangular spiral $r = \lambda e^{-k\theta}$.

Solution

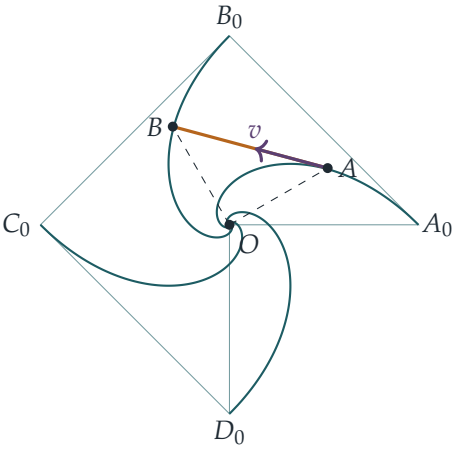
The dogs stay on a square

Take the origin O at the centre of the square and label the dogs anticlockwise, so that B is a quarter-turn anticlockwise from A, C from B, and so on.

Rotate the whole picture through $\frac{\pi}{2}$ about O. The starting positions of A, B, C, D go to the starting positions of B, C, D, A, and the rule "run at speed v straight at the next dog" is the same rule for every dog. So the rotated motion is also a motion obeying the same rules from the same starting configuration. The motion from a given start is unique (the velocities are smooth functions of the positions while the dogs are apart), so the rotated motion is the actual motion. In other words, at every instant B's position is A's position rotated through $\frac{\pi}{2}$ about O. The same goes for C and D. So $ABCD$ is always a square centred at O.

The angle between the velocity and the radius

Let A be at (r, θ) in polar coordinates. Then B is at $(r, \theta + \frac{\pi}{2})$, so $OA = OB = r$ and $\angle AOB = \frac{\pi}{2}$. Triangle OAB is right-angled and isosceles, so $\angle OAB = \frac{\pi}{4}$.



The figure shows the four paths spiralling in from the starting corners A_0, B_0, C_0, D_0, with the dogs A and B at one instant. A's velocity (plum) points along AB.

A's velocity has size v and points along AB, which makes an angle $\frac{\pi}{4}$ with the inward radius AO. Splitting it into the radial component (outwards positive) and the transverse component (in the direction of increasing θ):

$$\dot{r} = -v\cos\frac{\pi}{4}, \qquad r\dot{\theta} = v\sin\frac{\pi}{4}.$$

The transverse component is positive because B lies anticlockwise of A, so A is pulled round anticlockwise.

Ask yourself Why is it legitimate to divide these two equations?

Because $r\dot{\theta} = v/\sqrt{2} > 0$ while the dogs are apart, θ is strictly increasing in time. So we can use θ in place of t as the parameter along the path, and by the chain rule

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\dot{r}}{\dot{\theta}} = \frac{-v/\sqrt{2}}{v/(\sqrt{2}\,r)} = -r.$$

Solving for the path

Since $r > 0$ we can separate variables: $\displaystyle\int \frac{\mathrm{d}r}{r} = -\int \mathrm{d}\theta$, so $\ln r = -\theta + c$, that is

$$r = \lambda \mathrm{e}^{-\theta}$$

with $\lambda = \mathrm{e}^{c}$ a constant. This is the required result. If we choose the initial line to be OA_0, so that $\theta = 0$ at the start, then λ is A's initial distance from O, namely $a/\sqrt{2}$ for a square of side a. With any other initial line λ just changes by a constant factor. (If we had labelled the dogs clockwise, measuring θ clockwise gives the same equation.)

Generalisation to n dogs

For n dogs at the vertices of a regular n-gon (labelled anticlockwise, each chasing the next) the same argument works with a rotation through $\frac{2\pi}{n}$. The dogs always form a regular n-gon centred at O. So if A is at (r,θ), the next dog B is at $(r,\theta+\frac{2\pi}{n})$.

Triangle OAB is isosceles with apex angle $\frac{2\pi}{n}$, so each base angle is

$$\angle OAB = \frac{1}{2}\left(\pi - \frac{2\pi}{n}\right) = \frac{\pi}{2} - \frac{\pi}{n}.$$

Resolving A's velocity as before,

$$\dot r = -v\cos\left(\frac{\pi}{2}-\frac{\pi}{n}\right) = -v\sin\frac{\pi}{n}, \qquad r\dot\theta = v\sin\left(\frac{\pi}{2}-\frac{\pi}{n}\right) = v\cos\frac{\pi}{n}.$$

Since $n \geq 3$, $\cos\frac{\pi}{n} > 0$, so again θ increases steadily and

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = -r\tan\frac{\pi}{n} \qquad \Longrightarrow \qquad r = \lambda\,\mathrm{e}^{-\theta\tan(\pi/n)}.$$

For $n=4$, $\tan\frac{\pi}{4}=1$ and we recover $r=\lambda\mathrm{e}^{-\theta}$. Each dog follows an equiangular (logarithmic) spiral, crossing every radius at the fixed angle $\frac{\pi}{2}-\frac{\pi}{n}$.

A nice by-product: $\dot r = -v\sin\frac{\pi}{n}$ is constant, so the dogs meet at O after time $r_0/(v\sin\frac{\pi}{n})$, where r_0 is the initial distance from O. In that time each dog runs the finite distance $r_0/\sin\frac{\pi}{n}$, even though $\theta\to\infty$ and it circles O infinitely often. (For the square this is a, the side length.)

Checks

- For $n=4$ the result is the special case $\tan\frac{\pi}{4}=1$ of the general formula.
- As $n\to\infty$, $\tan\frac{\pi}{n}\to 0$ and the path tends to a circle: each dog is chasing a neighbour almost alongside it on the circle.
- We simulated the chase numerically (with RK4 for $n=3,4,5,6,8$ dogs, starting at radius 1). The computed r agreed with $\mathrm{e}^{-\theta\tan(\pi/n)}$ to a relative error of 10^{-11}. For $n=4$ the time and path length to reach $r=0.02$ came out as $1.386 = 0.98\sqrt{2}$, as predicted by $(1-0.02)/\sin\frac{\pi}{4}$.

Common slips

- Writing down $\dot r$ and $r\dot\theta$ without first justifying that the dogs stay on a regular polygon. The symmetry argument is what makes the angle at A known.
- Using the interior angle of the polygon ($\frac{\pi}{2}$ for a square) instead of the angle between AB and the radius AO ($\frac{\pi}{4}$).
- Forgetting that the transverse velocity component is $r\dot\theta$, not $\dot\theta$.

Beyond STEP

This is a classic *pursuit problem*. The symmetry step is an instance of a general principle in differential equations: if the equations and the initial data are both invariant under a symmetry, then uniqueness of solutions forces the solution to be invariant too. Curves meeting every radius at a fixed angle are the equiangular spirals. In complex form they are $z = \mathrm{e}^{(-k+\mathrm{i})\theta}$, so the whole motion is a continuous "spiral similarity" of the plane. Needham's *Visual Complex Analysis* develops this geometric view, and Boyce and DiPrima's *Elementary Differential Equations* covers existence and uniqueness.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.